

Chapter 3

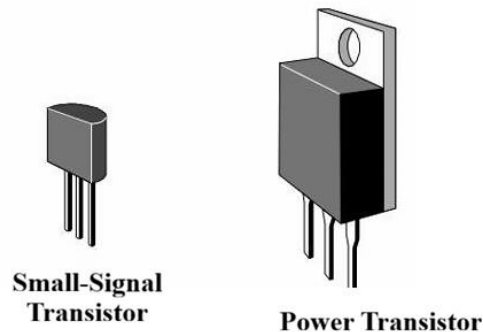
Bipolar Transistor

3.1 Definition of a Bipolar Transistor

The transistor was first developed in 1948. It is composed of two junctions connected in series, very close to each other, with opposite polarity.

A bipolar transistor is constructed by placing three doped semiconductor layers: N, P, then N for an **NPN transistor**, or P, N, then P for a **PNP transistor**.

A small base current, I_B , controls a much larger collector current, I_C .



3.2 Structure of a Bipolar Transistor

A bipolar transistor has **three regions**.

- **Emitter**
- **Base**
- **Collector**

The **collector** is the largest region.
It is usually connected to the transistor case.

The **base** is the middle layer.
It is **very thin** and separates the emitter from the collector.

Between these regions, there are **two junctions**:

- **Base–Emitter junction**
- **Collector–Base junction**

Because of these two junctions, the device is often called a **junction transistor**.

The two junctions (like two diodes) share the **base region**.

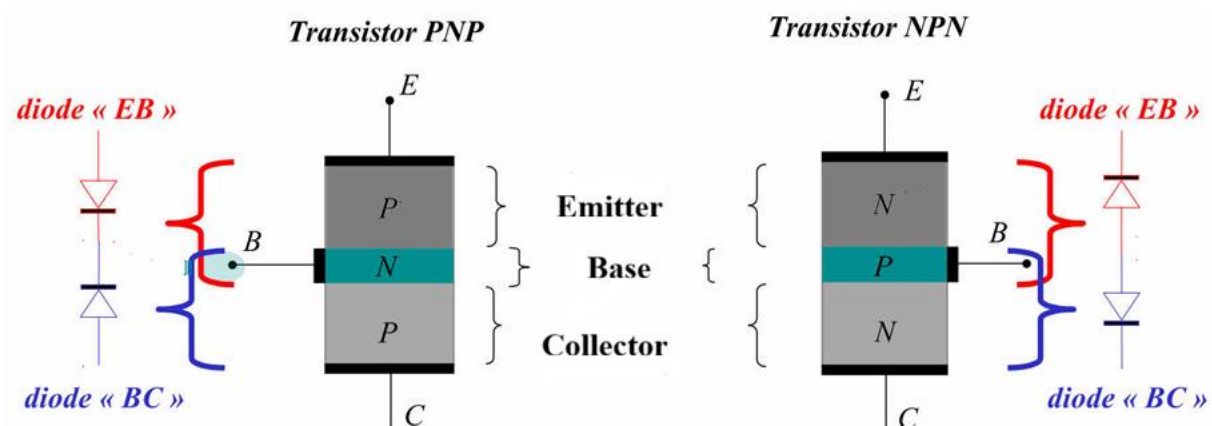


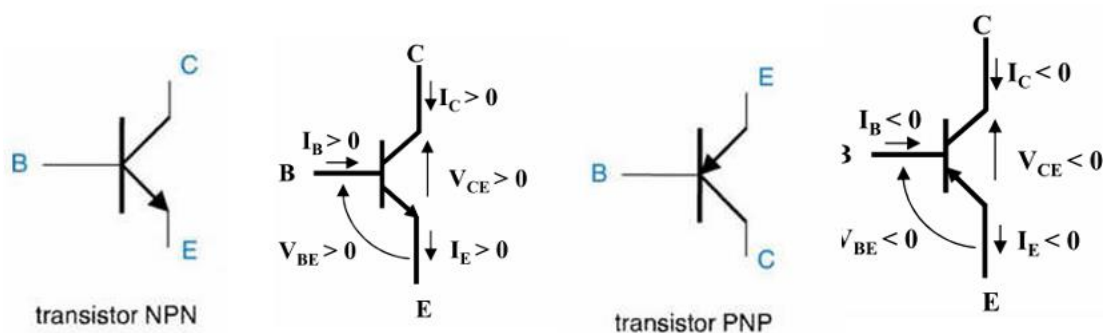
Figure 3.1: Basic Structure of NPN and PNP Bipolar Transistors.

3.2 Symbol

Schematically, transistors are represented as shown in **Figure 3.2**.

For each transistor, we define **six electrical quantities**:

- **Three currents**
- **Three voltages**



The PNP transistor works in the same way, but the currents and voltages are negative. The real current flows in the opposite direction to the arrows.

3.3 Fundamental current–voltage relationships

$$I_E = I_C + I_B$$

$$I_C = \beta I_B$$

β is the current gain, and it is very large.

$$I_C = \alpha I_E$$

As α is approximately equal to unity,

$$\beta \gg 1 \rightarrow I_C \gg I_B \rightarrow I_E = I_C + I_B \approx I_C$$

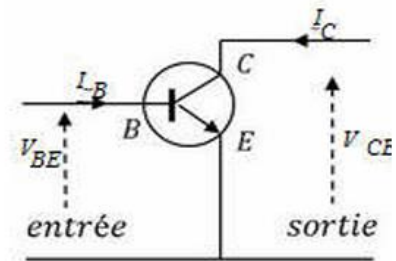
Silicium : $V_{BE} = 0.6V \text{ à } 0.7V$

Germanium : $V_{BE} = 0.2V \text{ à } 0.3V$

3.4 Static Characteristic Curves of the Transistor

Four quantities characterize the external behavior of the transistor:

- the input circuit: I_B as a function of V_{BE}
- the output circuit: I_C as a function of V_{CE}



Fundamentally, it is important to remember the following properties:

- The curve $I_B = f(V_{BE})$ corresponds to the operation of the base-emitter junction. It therefore represents the characteristic of a diode.

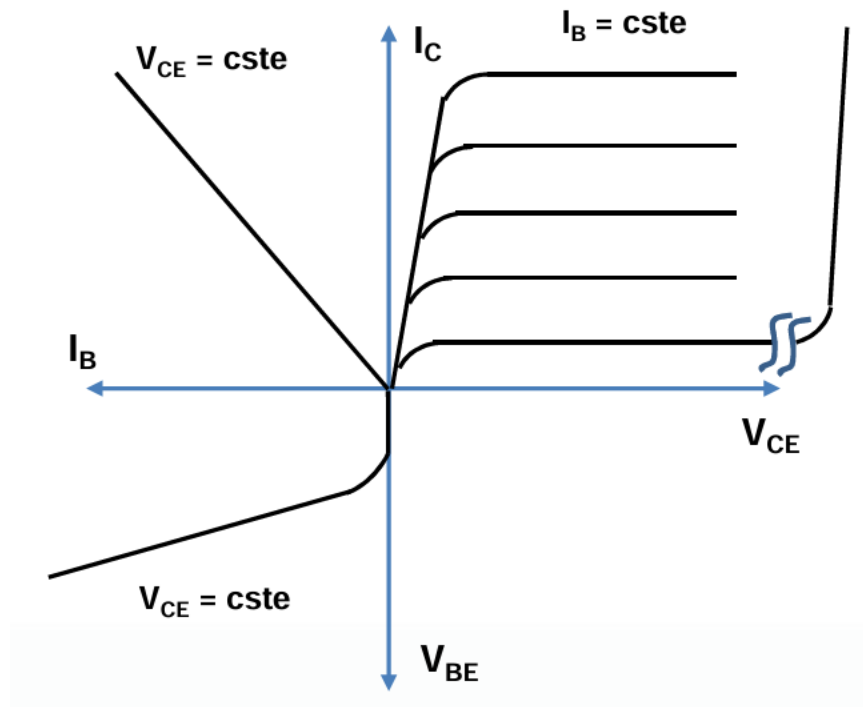
- The collector current I_C and the base current I_B are related by the fundamental relation:

$$I_C = \beta I_B.$$

The gain β is an intrinsic characteristic of the transistor; it is the current gain.

It is generally of the order of a few tens to a few hundreds, depending on the type of transistor.

- The function $I_C = f(V_{CE})$ depends on the value of I_B , which ultimately gives a family of characteristic curves.



3.5 Transistor Biasing

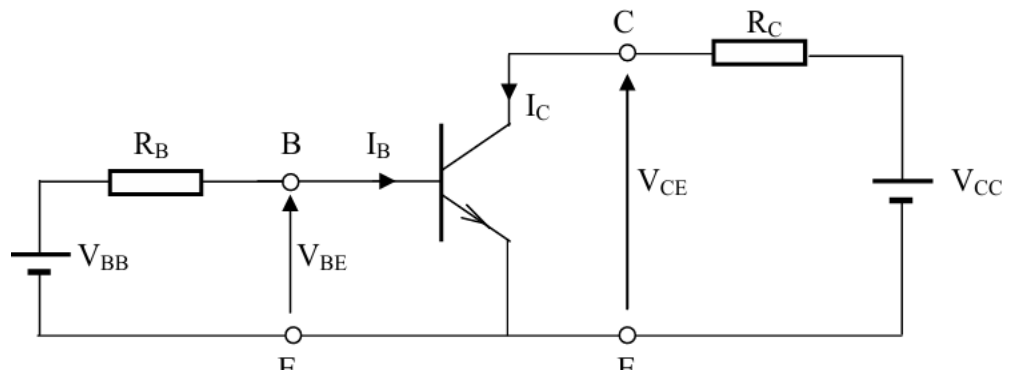
1) Definition

Biasing consists in defining the **static operating point** (quiescent point) of the transistor, characterized by the values V_{BE0} , I_{B0} , I_{C0} , and V_{CE0} .

There are different biasing methods.

2) Biasing using two voltage sources

a) Circuit diagram



b) Static Input Line (Driving Line)

It is defined by the equation:

$$V_{BE} = V_{BB} - R_B I_B$$

The intersection of this line with the **input characteristic of the transistor** gives the point (**I_{B0}** , **V_{BE0}**).

c) Static Load Line

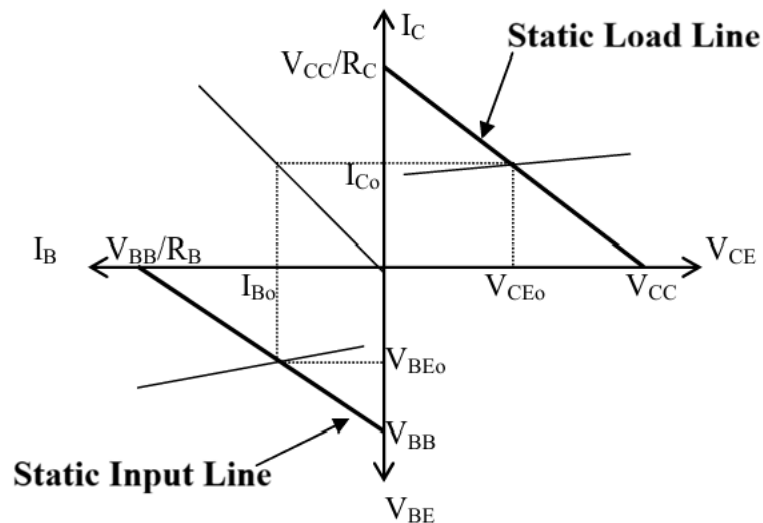
It is defined by the equation:

$$V_{CE} = V_{CC} - R_C I_C$$

The intersection of this line with the **output characteristic of the transistor** gives the operating point (**I_{C0}** , **V_{CE0}**).

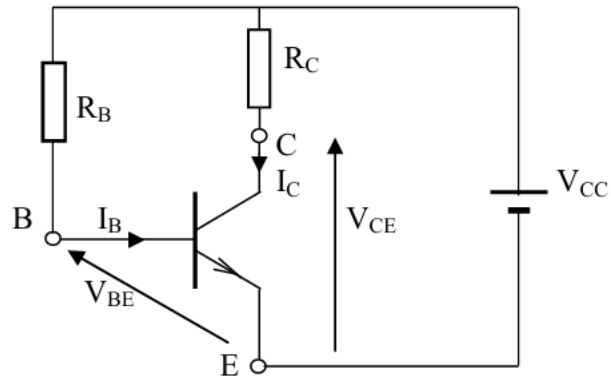
$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

The intersection of this line with the output characteristic of the transistor gives the point (**V_{CE0}** , **I_{C0}**).



3) Fixed Bias using Base Resistor

- Circuit diagram



- **Static Input Line (Driving Line)**

It is defined by the equation:

$$V_{BE} = V_{CC} - R_B I_B$$

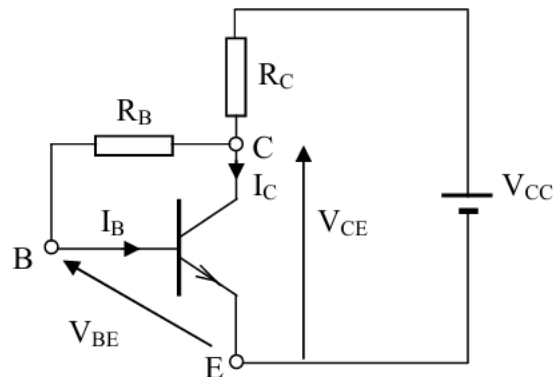
- **Static Load Line**

It is defined by the equation:

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

4 Biasing with a Resistor between Base and Collector

Circuit diagram



- **Static Input Line**

In this biasing case:

$$\begin{aligned} V_{BE} &= V_{CE} - R_B I_B \\ V_{CE} &= V_{CC} - R_C I_C \end{aligned}$$

With $I_C = \beta I_B$, we obtain:

$$V_{BE} = V_{CC} - (R_B + \beta R_C) I_B \quad \text{This is the equation of the **static input line** .}$$

- Static Load Line

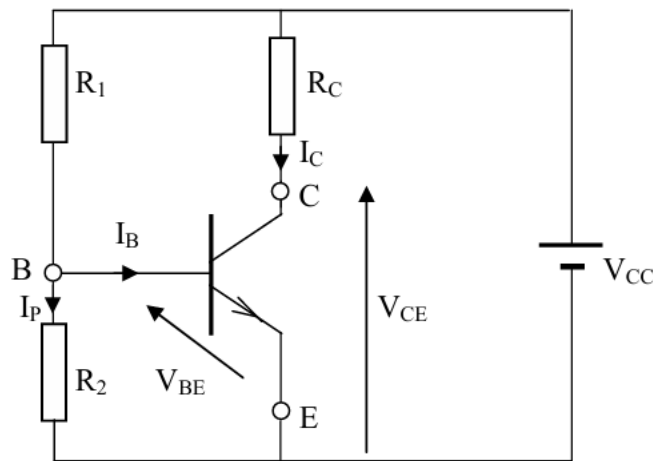
It is defined by the equation:

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

when I_B is neglected compared to I_C .

5 Biasing using a Base Voltage Divider (Resistor Network)

-Circuit diagram



- Static Input Line

In this biasing case, we can write:

$$I_P + I_B = \frac{V_{CC} - V_{BE}}{R_1}$$

with:

$$I_P = \frac{V_{CC}}{R_2}$$

Therefore, the equation of the **static input line** becomes:

$$V_{BE} = \frac{R_2}{R_1 + R_2} \cdot V_{CC} - \frac{R_1 \cdot R_2}{R_1 + R_2} \cdot I_B$$

(then it can be simplified depending on the circuit parameters)

- **Static Load Line**

It is defined by:

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$